MIL-G-12947A(MU)
20 MAY 1965
SUPERSEDING
MIL-G-12947 (MU)
5 AUGUST 1953

MILITARY SPECIFICATION

GRENADE, RIFLE, ILLUMINATING, M27 LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type or rifle grenade designated as Grenade, Rifle, Illuminating, M27.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-P-20342 — Primer, M45.

MIL-P-20464 — Pyrotechnics, Method of *Determination of* Candle-power and Color Value.

MIL-I-45607 —Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1234 — Pyrotechnics: Sampling, Inspection and Testing.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

76-1-1400 —Box, Packing, Ammunition for Rifle, Grenade, Illuminating, M27, Assembly, Details, and Packing.

76-1-1402 — Ammunition Container, Metal, for

FSC 1345

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Grenade, Rifle, Illuminating, M27, Assembly, Details, and Packing.

82–0–204 — Grenade, Rifle, Illuminating, M27, Assembly and Marking Diagram.

82–1–112 — Grenade, Rifle, Illuminating, M27, Illuminant Assembly and Details.

82–1–112A — Grenade, Rifle, Illuminating, M27, Illuminant Assembly, List of Parts and List of Specs and STD's.

PUBLICATIONS

ORDNANCE CORPS

IEL–7256768 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Grenade. The grenade shall comply with all requirements specified on drawing (dwg.) 82–0–204 and with all requirements specified in applicable specifications.

3.3 Moisture content.

3.3.1 *Illuminant composition.* The moisture content of the illuminant composition shall be 0.1 percent, maximum (max.) when tested as specified in 4.3.1.

3.3.2 *First fire composition.* The moisture content of the first fire composition shall be 0.1 percent, maximum, when tested as specified in 4.3.2.

3.4 Proving ground. The grenade shall comply with the following requirements when tested as specified in 4.3.3.

3.4.1 There shall be no rupturing of the stabilizer tube.

3.4.2 There shall be no premature functioning during launching or in flight before impact.

3.4.3 There shall be visible evidence of functioning at impact.

3.5 Workmanship. All parts and assemblies shall be loaded and assembled in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

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4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Statements of findings on all composition, powder and associated ingredients, and certificates of conformance on all remaining material when such material is covered by Government specifications.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of grenades loaded and assembled in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Lots shall be submitted for inspection in accordance with MIL–STD–105 (or MIL–STD–1235, when applicable). Each lot shall contain:

(a) Loaded illuminant assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and specification revision, of one loading lot interfix number.

(b) Stabilizer assemblies of one lot interfix number from one manufacturer.

(c) Loaded M45 Primers of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL–STD–1235 may be used, for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans

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may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Illuminant assembly (see dwg. 82–1–112).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Quickmatch or charge missing	Visual	01001
102.	Container damaged	Visual	01002
Minor:	AQL 1.50 percent		
201.	Length, max.	Gage	01003
202.	Diameter, max.	Gage	01004
203.	Cover assembly insecure	Manual	01005
204.	Evidence of poor workmanship (see 3.5)	Visual	01006

4.2.2.2 Grenade assembly (see dwg. 82–0–204).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Safety pin missing	Visual	02001
Major:	AQL 0.40 percent		
101.	Pettman cement missing or insufficient (prior to assembly)	Visual	02002
Minor:	AQL 2.50 percent		
201.	Length, max.	Gage	02003
202.	Diameter, max.	Gage	02004
203.	Fins distorted, loose, or bent	Visual-Manual	02005
204.	Marking incomplete, illegible or missing	Visual	02006
205.	Evidence of poor workmanship (see 3.5)	Visual	02007

4.2.2.3 Packing box and contents, prior to closing (see section 5).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Metal container damaged	Visual	03001
Minor:	AQL 0.65 percent		
201.	Marking incomplete, illegible or missing	Visual	03002

4.2.2.4 Sealed wood packing box (see section 5).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Hardware missing, loose or damaged	Visual-Manual	04001
202.	Board broken or split	Visual	04002
203.	Marking incomplete, illegible or missing	Visual	04003

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4.2.3 *Testing.*

4.2.3.1 *Moisture content of illuminant composition (see 3.3.1) Major defect—Code No. 05001.* The contractor shall provide adequate controls to assure compliance of each batch of composition with the applicable requirement. As a minimum, the contractor shall select one sample of the illuminant composition representative of each 8 hours production of grenades. If the moisture content of the sample fails to comply with the applicable requirement, the illuminant composition represented by the sample shall be rejected. If the illuminant composition with excessive moisture content has been used in loading, all assemblies containing the composition with excessive moisture content shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 *Moisture content of first fire composition (see 3.3.2) Major defect—Code No. 06001.* The contractor shall provide adequate controls to assure compliance of each batch of composition with the applicable requirement. As a minimum, the contractor shall select one sample of the first fire composition representative of each 8 hours production of grenades. If the moisture content of the sample fails to comply with the applicable requirement, the first fire composition represented by the sample shall be rejected. If the first fire composition with excessive moisture content has been used in loading, all assemblies containing the composition with excessive moisture content shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.3 *Determination of first fire composition and illuminant composition (see dwg. 82–1–112A)—Major defect—Code No. (see table I).* A sample quantity shall be selected from each batch for these determinations. If the sample fails to meet the requirements specified, the batch shall be rejected. The tests shall be performed as specifiel in 4.3.4 anl 4.3.5.

TABLE I

Composition	Code No.
First fire	07001
Illuminant	08001

4.2.3.4 *Proving ground.*

4.2.3.4.1 *Initial production—Code No. (see table II).* A sample of 125 grenades shall be selected from each lot for this test beginning with the first lot produced and continuing until three consecutive lots have passed the acceptance criteria specified herein. The lot shall be rejected is six or more grenades burn for 40 seconds or less or if eight or more grenades burn less than 55 seconds but more than 40 seconds or if any grenade bursts prematurely or if any stabilizer ruptures.

TABLE II

Defect	Code No.
a. Grenade burns 40 seconds or less-Major defect (see dwg. 82–1–112)	09001
b. Grenade burns less than 55 seconds but more than 40 seconds — Minor defect (see dwg. 82–1–112)	09002
c. Stabilizer tube ruptures — Critical defect (see 3.4.1)	09003
d. Grenade bursts prematurely — Critical defect (see 3.4.2)	09004

4.2.3.4.2 *Production testing—Code No. (see table II).* After three consecutive lots have met the criteria of 4.2.3.4.1, the sampling plan for this test shall be in accordance with MIL–STD–105 using code letter H (single sampling) and an AQL of 1.50 percent for defect (a) and an AQL of 2.50 percent for defect (b).

4.2.3.5 *Check test for possible deterioration of M45 primer (see Specification MIL–P–20432)—Code No. 10001.* If the total elapsed time between the original acceptance of the M45 primer lot and the assembly of that lot into the loaded grenades exceeds two years, or if the primers have been subjected to ad-

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verse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL–P–20342 immediately before the M45 primer lot is assembled into the grenades. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the grenades (see 6.4).

4.2.3.6 Static test of illuminant assembly for candlepower and burning time.

4.2.3.6.1 *Initial production—Code No. (see table III).* A sample of 125 grenades shall be selected from each lot for this test beginning with the first lot produced and continuing until three consecutive lots have passed the acceptance criteria specified herein. The lot shall be rejected if six or more assemblies burn for 40 seconds or less, or if eight or more assemblies burn less than 55 seconds but more than 40 seconds, or if six or more assemblies fail the applicable requirement for candlepower.

TABLE III

Defect	Code No.
a. Assembly burns 40 seconds or less — Major defect — (see dwg. 82–1–112)	11001
b. Assembly burns less than 55 seconds but more than 40 seconds —Minor defect — (see dwg. 62–1–112)	11002
c. Candlepower below minimum required — Major defect — (see dwg. 82–1–112)	11003

4.2.3.6.2 *Production testing—Code No. (see table III).* After three consecutive lots have met the criteria of 4.2.3.6.1, the sampling plan for this test shall be in accordance with MIL–STD–105 using code letter H (single sampling) and an AQL of 1.50 percent for defects (a) and (c) and an AQL of 2.50 percent for defect (b).

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–7256768 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Moisture content of illuminant composition.* The Karl Fischer Method as stated in MIL–STD–1234 shall be used for this test. A fifty gram sample plus or minus 0.1 gram shall be added to a 500 ml. volumetric flask containing approximately 300 to 400 ml. of methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents swirled cautiously for several min-

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utes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 ml. volumetric flask shall be filled up to the 500 ml. mark with methanol and swirled again. A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml. aliquot of the clear supernatant liquid shall be withdrawn and immediately transferred to the standard titration vessel containing approximately 100 ml. of methanol which has just been titrated to the preliminary end point as described in MIL-STD-1234, Method 101.2 paragraph 5.1. The final end point shall be reached in 3.5 minutes in the manner described in MIL-STD-1234, Method 101.2, paragraph 5.4. A 50 ml. aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

$$\text{Percent water} = \frac{(VR\text{-}S)\quad(V'R\text{-}S')\quad(100)}{W}$$

where:

F = grams of water per ml. of standard water in methanol solution.

V = ml. of Karl Fischer reagent added to the sample.

V' = ml. of Karl Fischer reagent added to the blank.

S = ml. of standard water in methanol solution for back titration of sample.

R = ml. of standard water in methanol solution per ml. of Karl Fischer reagent.

S' = ml. of standard water in methanol solution for back titration of blank.

W = sample weight in grams.

4.3.2 *Moisture content of first fire composition.* This test shall be performed as specified in MIL-STD-1234, method 101.2. Methyl alcohol shall be used as the special solvent.

4.3.3 *Proving ground.* The grenade shall be launched at approximately 45 degrees elevation by means of an M1 Rifle and M7A3 Launcher using the M3 Grenade Cartridge for impact on medium to hard ground. The M3 Grenade Cartridge shall be from one lot and shall be selected from the cartridges packed for field use with the lot of grenades.

4.3.4 *Determination of Illuminant composition (Magnesium only)—Eudiometer Method.* Exactly 1.0000 gram of the crushed sample shall be transferred to a 600 ml. wide-mouthed, Erlenmeyer flask. Two hundred ml. of distilled water saturated with hydrogen shall be added. The Erlenmeyer flask shall be connected to the apparatus. The 3 — way stopcock shall be opened so that the flask and the gas burette are exposed to atmospheric pressure. The level of the liquid shall be adjusted in the gas burette to the zero point with the leveling bulb. The 3-way stopcock shall be turned so that it will only allow gas to flow from the Erlenmeyer flask to the gas burette. The leveling bulb shall be lowered and 25 ml. of 6:19 sulfuric acid shall be added from the 25 ml. burette. After the addition of the acid, the flask shall be agitated occasionally until the reaction is com-

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pleted. The reaction shall be considered complete when the volume of generated gas remains within 1 ml. for a period of 20 minutes. When the reaction is complete, the level of the gas burette shall be adjusted to that of the water leveling bulb and the volume of gas shall be read in the gas burette. If the evolved gas is less than is measurable by the burette, a known volume of water shall be added through the acid burette, until a readable volume is reached on the gas burette. The determination must be started and finished at the same temperature (approximately 25 degrees C). The percentage of magnesium in the sample shall be calculated as follows:

$$\text{Percentage of magnesium} = \frac{(A-B)\quad(C-D)\quad .03901}{W\ (273\ \text{plus}\ t)}$$

where:

A = volume read on gas burette in ml.

B = volume of acid or volume of acid plus water added from the burette of ml.

C = atmospheric pressure in mm.

D = vapor pressure of water in mm. at the temperature of the determination.

t = temperature at which the determination is made.

W = weight of the sample.

Note. It is to be noted that the equations for calculating the percentages of the various ingredients are based upon the assumption that the ingredients added are 100 pure. If, however, it is found desirable to base the percentage of the ingredients present on any other purity, a factor can be introduced into the calculation.

4.3.5 *Determination of first fire composition.*

4.3.5.1 *Barium Nitrate.* An accurately weighed pjortion or exactly 1.000 gram of the ground sample shall be transferred to a tared medium porosity sintered glass crucible. The contents of the crucible shall be extracted with ten 20-ml. portions of distilled water, stirring each portion and allowing it to remain in contact with the residue for approximately 1 minute before applying suction. The crucible and contents shall be transferred to an oven at 100 degrees to 105 degrees C. for 1 hour, cooled to room temperature in a desiccator and weighed. The crucible and contents shall be reserved for the tetranitrocarbazole determination. The percentages of barium nitrate in the mixture shall be calculated as follows:

$$\text{Percentage of barium nitrate} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible and contents.

W = weight of the sample.

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4.3.5.2 *Tetranitrocarbazole.* The contents of the crucible reserved from the barium nitrate procedure shall be extracted with three 20 ml. portions of hot acetone, stirring each portion and allowing it to remain in contact with the residue for approximately 1 minute before applying suction. The crucible and contents shall be transferred to an oven at 100 degrees to 105 degrees C. for 1 hour, cooled to room temperature in a desiccator and weighed. The crucible and contents shall be reserved for the silicon determination. The percentage of tetranitrocarbazole in the mixture shall be calculated as follows:

$$\text{Percentage of tetranitrocarbazole} = \frac{100A}{W}$$

where:

A = weight loss in the crucible and contents.

W = weight of the sample.

4.3.5.3 *Silicon.* The crucible and contents reserved from the tetranitrocarbazole procedure shall be transferred to a 250 ml. beaker, 100 ml. of a 50 percent potassium hydroxide solution (50 grams KOH/100 ml. water) shall be added, the beaker shall be covered with a watch glass and transferred to a steam bath. The beaker and contents shall be heated for 1 hour with occasional swirling, removed from the steam bath and allowed to cool to room temperature. The crucible shall be removed carefully from the beaker with the aid of forcept, washing the crucible thoroughly with approximately 50 ml. of distilled water. The contents of the beaker shall be filtered quantitatively using the removed crucible. The residue in the crucible shall be washed with ten 20 ml. portions of distilled water. The crucible and contents shall be transferred to an oven at 100 degrees to 105 degrees for 1 hour, cooled to room temperature in a desiccator and weighed. The crucible and contents shall be retained for the mixture and shall be calculated as follows:

$$\text{Percentage of silicon} = \frac{.9240A\ (100)}{W}$$

where:

A = loss in weight of the crucible and contents.

W = weight of sample.

The factor of .9240 represents a correction factor for the solubility in the 50 percent potassium hydroxide solution of binder composition and impurities in the zirconium hydride. This figure shall be used except in the event of non-compliance, dispute or question. In this event, the weight of binder and impurities dissolved in the 50 percent potassium hydroxide solution shall be determined and the correction factor to be used shall be calculated as follows:

Exactly 0.1500 gram of zirconium hydride, 0.2000 gram of silicon and 0.0500 gram of binder composition of the specified grades shall be transferred to a medium porosity, sintered glass crucible. The procedure shall be exactly as in the above silicon determination and the correction factor shall be calculated as follows:

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$$\text{Correction factor} = \frac{0.2000}{A}$$

where:

A = loss in weight of the sample.

4.3.5.4 *Zirconium Hydride.* The crucible and contents reserved for the silicon procedure shall be transferred to a 250 ml. beaker. Fifty ml. of concentrated hydrochloric acid and 25 ml. of concentrated nitric shall be added. The beaker shall be covered with a watch glass and 50 ml. of concentrated sulfuric acid shall be added dropwise with caution. The beaker and contents shall be transferred to a hot plate, boiled gently until white sulfur trioxide fumes appear, and additional 75 ml. of concentrated sulfuric acid shall be added and the heating procedure shall be continued until a volume of approximately 25 ml. remains. During this heating period several drops of concentrated nitric acid shall be added cautiously under a watch blass from time to time until the binder composition is entirely decomposed, as evidenced by the permanent clarity of the solution. The solution shall be cooled to room temperature and sufficent distilled water to make a volume of approximately 100 ml. shall be added cautiously under a watch glass. The crucible shall be removed from the solution with the aid of forceps, washing it quantitatively into the beaker with approximately 50 ml. of distilled water. The solution shall be stirred thoroughly with a glass rod and allowed to cool to room temperature. The solution shall be filtered, using a Whatman No. 42 paper or substantial equal, collecting filtrate in a clean 400 ml. beaker. Any residue in the filter paper shall be washed with sufficient distilled water to make the total volume in the 400 ml. beaker approximately 200 ml. The filter paper shall be discarded and the beaker shall be cooled in an ice bath to 10 degrees plus or minus 2 degrees C. To the beaker in the ice bath shall be added with stirring, 20 ml. of freshly prepared cupferron solution (6 grams of cupferron shall be dissolved in 100 ml. of cold water and filtered). The precipitate shall be filtered using a Whatman No. 40 paper (or substantial equal) and washed with five 20 ml. portions of cold 1N (1:10) hydrochloric acid. The precipitate and paper shall be transferred to a tared crucible and dried in an oven at 100 degrees to 105 degrees C. for 2 hours. The crucible and contents shall be placed in a muffle furnace at 300 degrees to 400 degrees C. until complete charring of the paper has occurred and the rush of gases from decomposition of the organic matter has ceased. The temperature shall be increased to 900 degrees plus or minus 50 degrees C. and the ignition shall be continued for ½ hour. The crucible and contents shall be cooled to room temperature in a desiccator and weighed as zirconium oxide.

$$\text{Percentage of zirconium hydride} = \frac{75.65\ A}{W}$$

where:

A = gain in weight of the crucible.

W = weight of sample.

4.3.5.5 *Binder Composition.* The percentage of binder composition in the mixture shall be calculated by subtracting from 100 the combined percentage of barium nitrate, silicon, tetranitrocarbazole and zirconium hydride.

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4.3.6 *Static test of illuminant assembly for candlepower and burning time.* The grenade assembly shall be tested by removing the stabilizer and cover assembly and igniting the illuminant assembly, with burning end up, by means of quickmatch. Observation shall be made for compliance with the applicable requirements for candlepower and burning time. Determination of candlepower shall be made in accordance with MIL–P–20464.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The grenade shall be preserved, packaged and marked in accordance with drawing 76–1–1402.

5.2 Packing.

5.2.1 *Level A.* The packaged grenade shall be packed and marked in accordance with drawing 76–1–1400.

5.3 Data cards. Data cards shall be prepared for each lot as specified in MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the number column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF	— Contractor's design responsibility on Army format in accordance with MIL–D–45608.
CDCF	— Contractor's design responsibility on contractor format.
AI	— Army design.
ADMU	— Army design, mandatory for use.
CE	— Commercial equipment.
SCD	— Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into

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two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic full size prints of top quality with black lines on white background, and shall provide a legible print produced via the 35mm microfilm process. The contractor shall submit one rolled or flat set copy of the final approved design.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 **Cost of check test.** The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers

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or to the contractor assembling the grenades, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1345–A172

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DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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